



Basic Information

- Experiment No.: 16
- Date: 18/09/2026
- Title: Mobile Number Validation Using Custom Checked Exception and Built-in Exceptions
- Package Name: `record`
- Class Name: `PhoneNumber`



Objective / Aim

To design and implement a Java program that validates a mobile number using:

1. `NumberFormatException` for non-digit characters.
2. `ArrayIndexOutOfBoundsException` for numbers having more than 10 digits.
3. A custom checked exception `LengthNotSufficientException` for numbers having fewer than 10 digits.
4. `finally` block to close the `Scanner` object.



System Requirements

Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM
- Keyboard and monitor

Software Requirements

- Java Development Kit (JDK)
- IntelliJ IDEA / Eclipse / Command Prompt
- Operating System: Windows, Linux, or macOS

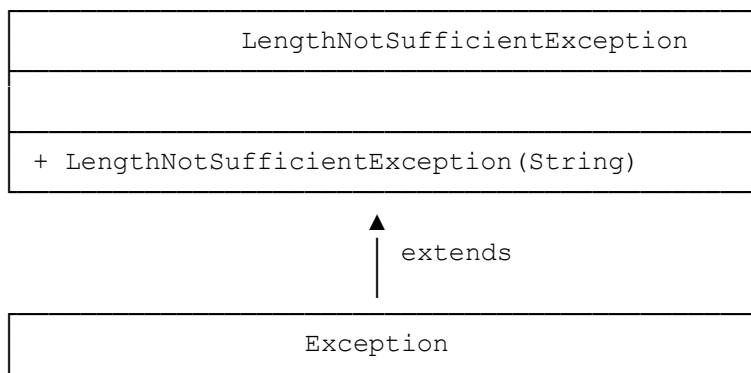


Algorithm / Logic Description

1. Start the program.
2. Create a custom checked exception named `LengthNotSufficientException` by extending the `Exception` class.
3. Read the mobile number from the user using `Scanner`.
4. Remove leading and trailing spaces using `trim()`.
5. Call the `validate()` method.
6. Check every character of the mobile number:
 - If any character is not a digit, throw `NumberFormatException`.
7. Check the length of the mobile number:
 - If the length is greater than 10, throw `ArrayIndexOutOfBoundsException`.
 - If the length is less than 10, throw `LengthNotSufficientException`.
8. If the length is exactly 10 and all characters are digits, display `Valid number`.
9. Handle the exceptions using separate `catch` blocks.
10. Execute the `finally` block and close the `Scanner`.
11. Stop the program.

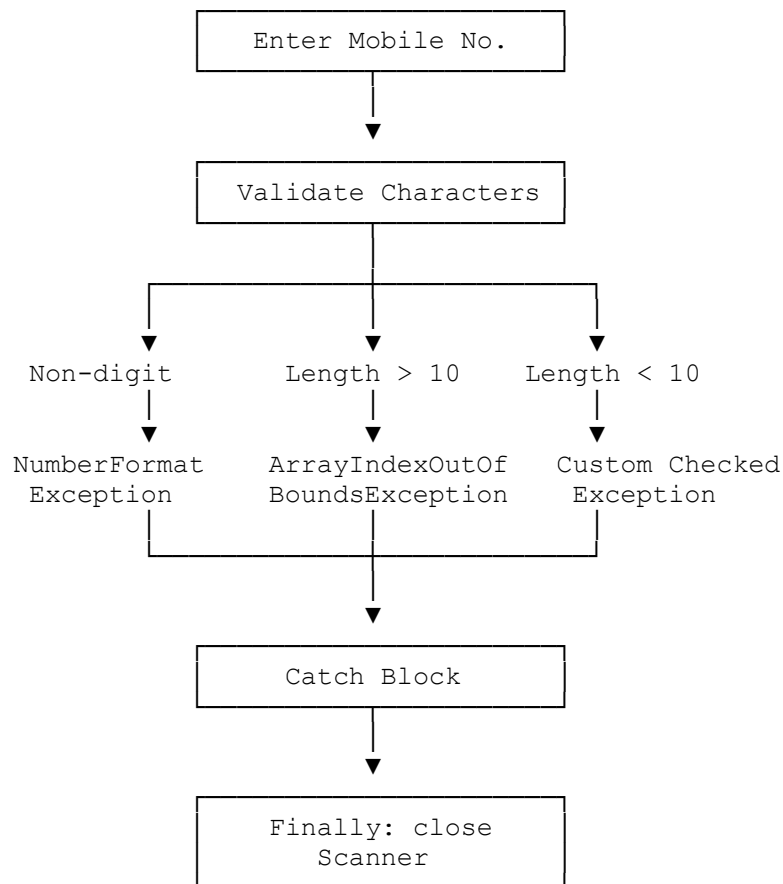


UML Class Diagram



PhoneNumber
<pre>+ validate(String): void + main(String[]): void</pre>

Exception Handling Flow



Source Code

```
import java.util.Scanner;

// custom checked exception
class LengthNotSufficientException extends Exception {
    LengthNotSufficientException(String message) {
        super(message);
    }
}
```

//PhoneNumber class

```
public class PhoneNumber {
```

// Validates the number and throws the appropriate exception

```
static void validate(String number)
```

```
    throws LengthNotSufficientException {
```

// Rule 1: every character must be a digit (0-9)

```
for (int i = 0; i < number.length(); i++) {
```

```
    if (!Character.isDigit(number.charAt(i))) {
```

```
        throw new NumberFormatException(
```

```
            "Invalid Mobile Number - NumberFormatException");
```

```
    }
```

```
}
```

// Rule 2: length greater than 10

```
if (number.length() > 10) {
```

```
    throw new ArrayIndexOutOfBoundsException(
```

```
        "Invalid Mobile Number - ArrayIndexOutOfBoundsException");
```

```
}
```

// Rule 3: length less than 10

```
if (number.length() < 10) {
```

```
    throw new LengthNotSufficientException(
```

```
        "Invalid Mobile Number - LengthNotSufficientException");
```

```
}
```

```
System.out.println("Valid number");
```

```
}
```

```
public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
```

```
    System.out.print("Enter Mobile Number: ");
```

```
    String number = sc.nextLine().trim();
```

```
    try {
```

```
        validate(number);
```

```
    }
```

```
    catch (NumberFormatException e) {           // non-digit character
```

```
        System.out.println(e.getMessage());
```

```
    }
```

```
    catch (ArrayIndexOutOfBoundsException e) { // more than 10 digits
```

```
        System.out.println(e.getMessage());
```

```
    }
```

```
    catch (LengthNotSufficientException e) { // fewer than 10 digits
```

```
        System.out.println(e.getMessage());
```

```
    }  
    finally {  
        sc.close();  
    }  
}  
}
```

Compilation & Execution Commands

Step 1: Compile the Program

Save the file as:

```
PhoneNumber.java
```

Compile using:

```
javac -d . PhoneNumber.java
```

Step 2: Execute the Program

```
java record.PhoneNumber
```



Output Screen

```
"C:\Program Files\Java\jdk-24\bin\ja  
Enter Mobile Number: 8341544080  
Valid number  
  
Process finished with exit code 0  
|
```